

# Quantitative Analysis of Media Bias and Stock Price Dynamics: The 2020 Shock

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**Abstract**—This paper studies how media reporting about large-cap firms changed around the 2020 recession and whether shifts in news stance predict firm-level stock outcomes. Using a large-scale NewsMTSC pipeline to compute daily stance scores from full-text news articles (2015–2025) for S&P 500 firms, we combine NLP-derived stance with firm-day controls and market variables to test structural shifts, causal responses, and predictive linkages. Our identification strategy centers on a pre/post shock design around January 2020, complemented by difference-in-differences, interrupted time series, and time-series VAR exercises. We document the data pipeline, relevance filtering, stance aggregation, and robustness checks used throughout the analysis.

## I. INTRODUCTION

The 2020 recession is a natural breakpoint for studying the interaction between media narratives and financial markets. COVID-era coverage expanded in volume and intensity, and a growing body of evidence documents a structural change in news-market dynamics during and after the shock. This project asks a simple, high-level question: did the tone of firm-specific media reporting shift after 2020, and if it did, did markets respond differently to that tone?

We approach the problem with a target-dependent stance framework rather than generic sentiment. NewsMTSC allows us to score whether an article is positive or negative toward a specific firm even when multiple firms are mentioned in the same text. That distinction is central to firm-level testing: a general sentiment score cannot distinguish positive coverage of one firm from negative coverage of another in the same article. Target-dependent stance makes it feasible to construct a daily, firm-level media stance signal and evaluate how it behaves across time and shocks.

The study is grounded in three observations from the project work so far. First, the 2020 shock is widely documented as a period in which media coverage and market behavior changed in a persistent way. Second, the data pipeline needs to be precise: naive keyword search introduces many false positives and inflates noise in the stance signal. Third, if stance is to be linked to returns in a credible way, the design must rule out

coverage volume effects, pre-trend differences, and spurious correlations driven by macro shocks.

We therefore build a two-stage pipeline: (i) relevance classification that filters articles to those that actually report on the target firm, and (ii) NewsMTSC stance scoring with confidence thresholds. With this signal we pose three research questions on structural shifts in stance and returns and the post-shock stance-return linkage. Our empirical strategy combines pre/post regressions, interrupted time series, difference-in-differences, VAR and Granger causality, and event studies to triangulate the stance-return channel.

The remainder of the paper proceeds as follows. Section II states the research questions and formal hypotheses. Section III describes the data pipeline. Section IV develops the methodology and estimating equations. Section V presents results for each RQ. Section VI addresses robustness, and Section VII concludes.

## II. RESEARCH QUESTIONS AND HYPOTHESES

Three hypotheses organize the empirical work. In each case the null corresponds to no effect; rejection constitutes a positive finding.

**RQ1: Stance shift.** Did the tone of firm-specific media coverage shift systematically after January 2020?

$$H_0^{(1)} : \beta = 0$$

$$H_1^{(1)} : \beta \neq 0$$

where  $\beta$  is the coefficient on  $\text{Post}_t$  in the two-way fixed-effects stance regression (Section IV).

**RQ2: Returns shift.** Did firm-idiosyncratic daily returns shift after January 2020 beyond market co-movement?

$$H_0^{(2)} : \beta = 0$$

$$H_1^{(2)} : \beta \neq 0$$

where  $\beta$  is the coefficient on  $\text{Post}_t$  in the two-way fixed-effects returns regression.

**RQ3: Stance-return predictability.** Does past media stance Granger-cause future returns, and did that relationship change post-2020?

$$H_0^{(3)} : a_1^{(2,1)} = \dots = a_p^{(2,1)} = 0$$

$$H_1^{(3)} : \exists \ell \text{ s.t. } a_\ell^{(2,1)} \neq 0$$

where  $a_\ell^{(2,1)}$  are the lagged-stance coefficients in the return equation of the VAR.

### III. DATA AND SIGNAL CONSTRUCTION

#### A. Corpus and Coverage

We begin with a large corpus of scraped full-text news articles (2015–2025). The core sample focuses on large-cap firms that are consistently observable across time, namely the S&P 500. Articles are stored with publication, timestamp, URL, title, first paragraph, and full body text. We de-duplicate on titles and keep a separate record of filtered-out content to preserve traceability.

To avoid severe imbalance across firms and time, we use a stratified design: for each year we compute article counts per firm, take the minimum feasible count across firms, and sample to match that count. This preserves coverage across firms while avoiding a signal driven by a handful of highly covered companies.

#### B. Sampling and Stratification (Facts and Figures)

The starting panel contains 4,706,589 firm-article links across the top firms and years 2015–2025 (Table I). This raw panel is highly skewed: some firms dominate the coverage while others have sparse years.

We therefore impose two sampling constraints. First, we filter to articles whose titles explicitly contain the company name and de-duplicate by normalized title per company. Second, we apply a minimum annual coverage threshold: a company is retained if it has at most three years with  $\leq 3,000$  articles; companies failing this are dropped. Table II shows the full construction funnel from raw panel to analysis sample.

Figure 1 shows the long-tail distribution of article counts per company, motivating stratification. Figure 2 summarizes the top-coverage firms in the retained panel.

#### C. Relevance Filtering and Annotation

Keyword matching is not enough for firm targeting (e.g., “Apple” is not always the company). We therefore apply a two-stage relevance pipeline: (i) title-based pre-filtering that retains only articles with the company name in the title (company names, not tickers), and (ii) a supervised relevance classifier trained on manual annotations. The held-out test set for manual annotation is kept completely separate from training.

Our base classifier is DeBERTa-v3 fine-tuned for binary classification using article titles. The first model, trained on 750 labeled articles, achieved high recall but insufficient precision. We then expanded the training data via semi-supervised mining: running the model on a large unlabeled pool and adding only high-confidence predictions ( $p_{rel} \geq 0.92$

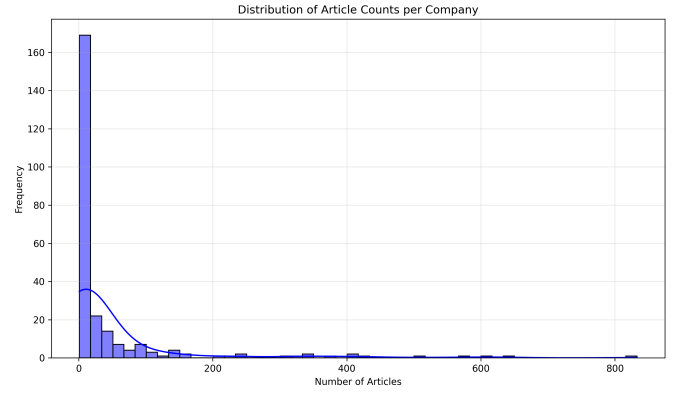


Fig. 1. Distribution of article counts per company. The long tail motivates stratified sampling to avoid coverage-driven bias.

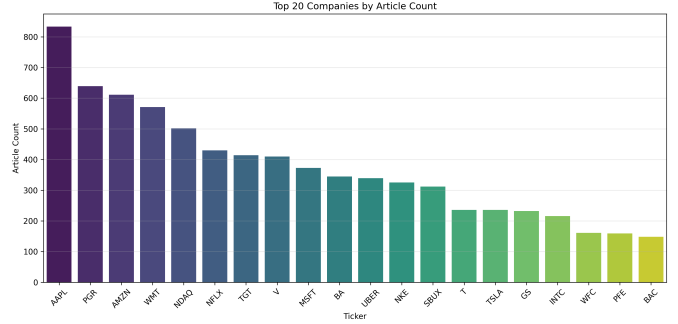


Fig. 2. Top 20 firms by article count in the retained sample.

for relevant,  $p_{rel} \leq 0.05$  for irrelevant). This expanded set increases precision and reduces false positives. An ensemble of model variants further improves precision, and the operating threshold is selected via a sweep to meet a target of precision  $\geq 0.90$ .

#### D. Stance Scoring

Stance is computed using the NewsMTSC model for target-dependent stance. For each qualifying article we obtain probabilities for positive, negative, and neutral stance. The baseline specification retains only high-confidence outputs ( $\text{prob } \hat{c} = 0.9$ ), and we report sensitivity to lower thresholds (0.6 and 0.0). Daily firm-level stance is the average across articles on that day:

$$\text{stance}_{i,t} = \frac{1}{N_{i,t}} \sum_{a \in A_{i,t}} (p_a^+ - p_a^-) \quad (1)$$

where  $A_{i,t}$  is the set of high-confidence articles for firm  $i$  on day  $t$  and  $N_{i,t}$  is the count. We also construct article volume and coverage mix covariates to separate tone shifts from coverage shifts.

#### E. Market Alignment and Controls

The stance series is aligned to firm-day market outcomes: daily returns, S&P 500 index returns, and VIX. We also log-normalize article volume in regressions. This alignment yields the panel used across the RQ1–RQ3 specifications.

TABLE I  
YEARLY ARTICLE TOTALS IN THE STARTING PANEL (2015–2025)

| Year           | 2015    | 2016    | 2017    | 2018    | 2019    | 2020    | 2021    | 2022    | 2023    | 2024    | 2025    |
|----------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| Total articles | 570,999 | 625,582 | 652,741 | 525,259 | 559,027 | 363,962 | 291,144 | 260,115 | 204,281 | 204,007 | 449,472 |

TABLE II  
SAMPLE CONSTRUCTION FUNNEL

| Stage                                    | Firms | Firm-article links |
|------------------------------------------|-------|--------------------|
| Raw panel (2015–2025)                    | 30    | 4,706,589          |
| Title deduplication                      | 30    | —                  |
| Coverage filter ( $\leq 3$ sparse years) | 22    | —                  |
| Final stratified panel                   | 22    | 483,514            |

#### IV. METHODOLOGY

##### A. Identification and Panel Design

Both RQ1 and RQ2 use a pre/post design on a firm–day panel. The treatment indicator

$$\text{Post}_t = 1[t \geq 2020-01-01] \quad (2)$$

partitions the sample into a pre-period (2015–2019) and post-period (2020–2025). Firm fixed effects  $\alpha_i$  absorb permanent cross-firm heterogeneity; time fixed effects  $\delta_t$  absorb common shocks at the day or month level depending on the outcome. The stance outcome is the signed daily mean,

$$\text{bias}_{it} = \frac{1}{|\mathcal{A}_{it}|} \sum_{a \in \mathcal{A}_{it}} (p_a^+ - p_a^-), \quad (3)$$

and the returns outcome is the log return  $\text{return}_{it} = \ln P_{it} - \ln P_{i,t-1}$ . Both equations are estimated by within-transformation OLS with HC1-robust standard errors; day-clustered standard errors are additionally reported for RQ1.

##### B. RQ1: Stance Regression

$$\text{bias}_{it} = \alpha_i + \delta_t^{\text{day}} + \beta \text{Post}_t + \gamma \text{vol}_{it} + \varepsilon_{it} \quad (4)$$

Firm effects  $\alpha_i$  absorb each firm’s structural coverage level; daily time effects  $\delta_t^{\text{day}}$  absorb market-wide sentiment common to all firms on a given day. The volume control  $\text{vol}_{it} = |\mathcal{A}_{it}|$  is required because the mean of a noisy signal mechanically concentrates as more articles are averaged, so volume must be held fixed to isolate a genuine tone shift from a coverage shift. Under  $H_0^{(1)}$ ,  $\beta = 0$ .

##### C. RQ2: Returns Regression

$$\text{return}_{it} = \alpha_i + \delta_t^{\text{month}} + \beta \text{Post}_t + \gamma_1 \text{sp500\_ret}_t + \gamma_2 \text{vix}_t + \varepsilon_{it} \quad (5)$$

Monthly time effects  $\delta_t^{\text{month}}$  absorb common market-level shocks at the monthly frequency, which is the resolution at which macro regime changes operate. The S&P 500 return and VIX are included as daily controls within each month. Under  $H_0^{(2)}$ ,  $\beta = 0$ .

##### D. RQ3: VAR and Granger Causality

For the dynamic analysis we estimate a reduced-form bivariate VAR of order  $p$  on the endogenous vector  $\mathbf{y}_t = (\Delta \text{bias}_t, r_t)^\top$ :

$$\mathbf{y}_t = \mathbf{c} + \sum_{\ell=1}^p \mathbf{A}_\ell \mathbf{y}_{t-\ell} + \mathbf{u}_t \quad (6)$$

where  $\Delta \text{bias}_t$  is the first-differenced weekly bias index (required for stationarity; see Section VI) and  $r_t$  is the weekly log return. The  $H_0^{(3)}$  restriction is  $a_1^{(2,1)} = \dots = a_p^{(2,1)} = 0$ , tested via a joint Wald  $F$ -test with Newey–West HAC standard errors.

#### V. RESULTS

##### A. RQ1: No Significant Aggregate Stance Shift

Table III(a) reports estimates from Eq. (4) on  $n = 63,200$  firm–day observations (26 firms, 4,018 days). We fail to reject  $H_0^{(1)}$ : the post-shock coefficient is  $\hat{\beta} = +0.0141$  with HC1 SE = 0.0136 ( $p = 0.30$ ); day-clustered inference is nearly identical (SE = 0.0134,  $p = 0.29$ ), confirming that within-day cross-firm dependence is not driving the result. Article volume is strongly negative ( $-0.00097^{***}$ ); heavier-coverage firm-days regress toward neutral tone, validating the volume control. The within  $R^2 = 0.0017$  reflects the well-known dominance of idiosyncratic article noise at the firm-day level.

The null result is informative. A joint  $F$ -test of the 4,017 daily effects is strongly significant ( $F = 1.135$ ,  $p = 1.1 \times 10^{-8}$ ), meaning common daily shocks drive most of the time variation in stance. Once those shocks are absorbed by  $\delta_t^{\text{day}}$ , the static Post step carries almost no residual identifying variation—the day fixed effects effectively span it. We therefore fail to detect a systematic within-firm stance shift beyond what the broader news environment already explains.

##### B. RQ2: No Significant Firm-Idiosyncratic Return Shift

Table III(b) reports estimates from Eq. (5) on  $n = 62,667$  firm–day observations (26 firms; pre-period mean returns tightly clustered, range 0.0019, SD 0.0004). We fail to reject  $H_0^{(2)}$ : once monthly time effects absorb common market-regime variation, the post-shock coefficient is  $\hat{\beta} = -0.0018$  (SE = 0.0142,  $p = 0.90$ ). The S&P 500 return regressor is absorbed by the month fixed effects ( $\hat{\gamma}_1 \approx 0$ ,  $p = 0.90$ ), and VIX is similarly uninformative within month ( $p = 0.93$ ). The within  $R^2 \approx 0$  follows directly: with the dominant systematic factor (index returns) loaded entirely onto the monthly FE, the regressors have negligible residual explanatory power.

Together, RQ1 and RQ2 establish a clean null baseline: after properly absorbing common daily and monthly shocks,

TABLE III

REGRESSION RESULTS FOR RQ1 (STANCE SHIFT, EQ. 4) AND RQ2 (RETURNS SHIFT, EQ. 5). BOTH SPECIFICATIONS FAIL TO REJECT THE NULL AT CONVENTIONAL LEVELS ONCE COMMON DAILY (RQ1) AND MONTHLY (RQ2) SHOCKS ARE ABSORBED.

| (a) RQ1 — Daily Stance              |          |         |         |
|-------------------------------------|----------|---------|---------|
| Firm + day FE, HC1 SE               |          |         |         |
| Variable                            | Coef.    | SE      | $p$     |
| Post ( $\hat{\beta}$ )              | +0.01413 | 0.01362 | 0.300   |
| Article volume                      | −0.00097 | 0.00008 | < 0.001 |
| $n = 63,200$ ; 26 firms; 4,018 days |          |         |         |
| Within $R^2 = 0.0017$               |          |         |         |

| (b) RQ2 — Daily Returns |             |         |       |
|-------------------------|-------------|---------|-------|
| Firm + month FE, HC1 SE |             |         |       |
| Variable                | Coef.       | SE      | $p$   |
| Post ( $\hat{\beta}$ )  | −0.00184    | 0.01424 | 0.897 |
| S&P 500 return          | $\approx 0$ | —       | 0.902 |
| VIX                     | +0.00061    | 0.00654 | 0.925 |
| $n = 62,667$ ; 26 firms |             |         |       |
| Within $R^2 \approx 0$  |             |         |       |

neither media stance nor firm-level returns show a detectable aggregate level shift around January 2020. This motivates the dynamic question in Section VI—whether the *lead-lag relationship* between stance and returns changed post-shock, even if the marginal distributions did not.

## VI. VECTOR AUTOREGRESSION AND GRANGER CAUSALITY ANALYSIS

The regressions in the previous section ask whether the level of each series shifts after 2020. They are silent on the dynamic relationship between media stance and returns: whether past stance helps predict future returns, whether returns feed back into coverage, and whether that lead-lag structure itself changed after the shock. We answer these questions with a vector autoregression (VAR) and Granger-causality framework, supported by an eight-stage data pipeline that constructs a clean, stationarity-validated weekly panel before any dynamic model is fit. We move to a weekly frequency for this analysis because daily firm-level stance is sparse and noisy for several firms, whereas weekly aggregation yields near-complete coverage (median 99.8% of weeks) while retaining enough observations ( $\approx 522$  weeks per firm) for lag-rich models.

### A. The VAR Model

For a firm with endogenous vector  $\mathbf{y}_t = (\Delta \text{bias}_t, r_t)^\top$ , where  $\Delta \text{bias}_t$  is the first difference of the weekly bias index and  $r_t$  is the weekly log return, a reduced-form VAR of order  $p$  is

$$\mathbf{y}_t = \mathbf{c} + \sum_{\ell=1}^p \mathbf{A}_\ell \mathbf{y}_{t-\ell} + \mathbf{u}_t, \quad (7)$$

with coefficient matrices  $\mathbf{A}_\ell \in \mathbb{R}^{2 \times 2}$  and innovations  $\mathbf{u}_t$  having covariance  $\Sigma$ . The off-diagonal entries of the  $\mathbf{A}_\ell$  carry the cross-dynamics: a nonzero (1, 2) entry means past returns move current stance, and a nonzero (2, 1) entry means past stance moves current returns. The VAR is the minimal model that treats both variables as endogenous and lets the data, rather than an a priori causal ordering, reveal the direction of predictability, which is precisely what a media-market feedback question requires.

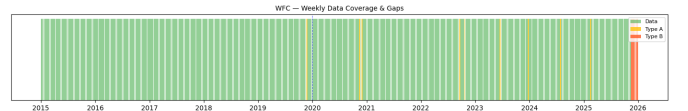


Fig. 3. Weekly data-coverage and gap map for Wells Fargo (WFC), the lowest-coverage retained firm. Short and medium gaps are imputed; the series still clears the  $\geq 60\%$  coverage threshold.

### B. Stage 1–4: Building the Weekly Panel

The pipeline begins with a *data inventory* (Stage 1) that audits, for every candidate firm, article counts, coverage, and the share of zero-article days, and flags sparse firms ( $< 30\%$  trading-day coverage) for exclusion. This audit motivates the move to a weekly frequency and the retention of sixteen well-covered firms. Stage 2 constructs the *weekly bias index* by aggregating signed article stances into ISO weeks and verifies distributional quality (mean, skew, kurtosis, and the incidence of single-article weeks). Stage 3 performs *gap analysis and imputation*: weekly gaps are classified by length, with short Type-A gaps (1–2 weeks) forward-filled and medium Type-B gaps (3–8 weeks) linearly interpolated, while long Type-C gaps are left missing; only 58 observations require imputation and all sixteen firms clear a  $\geq 60\%$  coverage threshold with no gap exceeding eight weeks. Crucially, both an imputed and an observed-only series are retained so that every downstream test can be re-run to confirm results are not artifacts of imputation. Figure 3 illustrates the coverage map for the sparsest retained firm.

Stage 4 constructs the *returns series* as weekly log returns from Monday-anchored ISO weeks and aligns them to the bias index. Log returns are used because they are additive across time and approximately symmetric, which suits a linear dynamic model. Quality checks flag extreme weekly moves ( $> \pm 15\%$ , concentrated in the March 2020 crash window) for later winsorization, and every firm clears the minimum of 100 aligned weeks, yielding a merged panel of 8,090 firm-week rows.

### C. Stage 5: Stationarity and Cointegration

A VAR in levels is only valid if the series are stationary or jointly cointegrated; otherwise the regression is spurious.

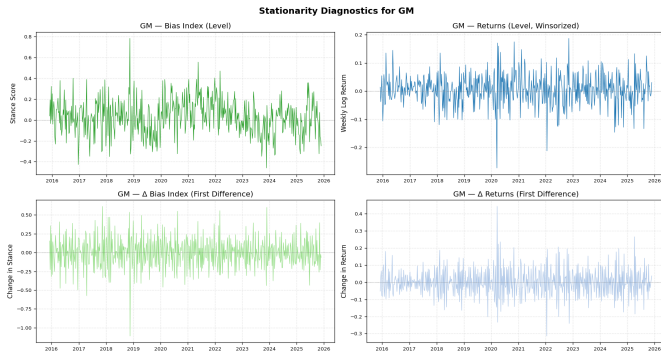


Fig. 4. Stationarity diagnostics for General Motors (GM): the bias index and weekly return in levels (left) and first differences (right), with ADF and KPSS verdicts driving the integration-order classification.

Each series is classified by combining the Augmented Dickey–Fuller (ADF) test, whose null is a unit root, with the KPSS test, whose null is stationarity. A series is called  $I(0)$  when ADF rejects ( $p < 0.05$ ) and KPSS does not,  $I(1)$  in the opposite case, and “conflicting” (a likely structural break) when both reject. As expected for financial data, weekly log returns are  $I(0)$  for essentially every firm, while the bias index is predominantly  $I(1)$  or break-driven in levels but  $I(0)$  in first differences. This mixed integration order is the reason the VAR is specified on  $(\Delta \text{bias}_t, r_t)$  rather than on the levels: differencing the bias index renders it stationary and avoids a spurious levels regression against  $I(0)$  returns. A Johansen trace test detects a cointegrating relation only for Wells Fargo (trace = 340.2 against a 95% critical value of 15.5), which would otherwise call for a VECM; for the common  $(I(1), I(0))$  case cointegration is impossible, confirming the differenced specification. Figure 4 shows the level-versus-difference diagnostic for a representative firm.

#### D. Stage 6: Structural Break Testing

Because the central hypothesis concerns a regime change around 2020, we date breaks empirically rather than imposing the calendar. A Bai–Perron procedure, implemented by dynamic programming for a single change point under an  $L_2$  (mean-shift) cost, locates the largest structural shift in the mean of each weekly series. The estimated bias-index break dates serve as the `is_post_break` cut used to split the sample for the subsequent subsample Granger tests, and every firm yields a valid split with at least 60 observations on each side. Figures 5 and 6 show the data-driven breaks for two firms; several cluster near the COVID window while others reflect firm-specific events, which is itself evidence that a single imposed 2020 dummy would be too blunt.

#### E. Stage 7: VAR Specification

For each firm the VAR order is selected by estimating Eq. (6) for  $p = 1, \dots, 8$  and minimizing the Bayesian Information Criterion (BIC), which penalizes parameters more heavily than AIC and so favors parsimonious, well-identified models; when BIC selects zero lags we impose  $p = 1$  to



Fig. 5. Bai–Perron structural breaks for Visa (V). The return break (2020-03-23) coincides with the COVID crash; the bias break (2020-09-21) lags it.



Fig. 6. Bai–Perron structural breaks for Wells Fargo (WFC), the one firm with a cointegrating bias–return relation and significant full-sample causality.

retain a dynamic model. The selected orders range from 1 to 5 (modal value  $p = 3$ ). Three residual diagnostics certify each fit. Stability requires all roots of the companion matrix to lie outside the unit circle, equivalent to  $\det(\mathbf{I} - \sum_{\ell} \mathbf{A}_{\ell} z^{\ell}) \neq 0$  for  $|z| \leq 1$ ; every selected model is stable (minimum root  $> 1.28$ ). The Portmanteau test checks for residual autocorrelation, confirming the chosen lag is adequate for most firms. The Jarque–Bera test rejects normality everywhere, the familiar heavy-tail signature of financial data, which leaves VAR coefficients consistent but motivates robust inference; Engle’s ARCH test detects volatility clustering in several return residuals, reinforcing the use of HAC-robust standard errors in the causality stage.

#### F. Stage 8: Granger Causality

Granger causality operationalizes “does past  $X$  help predict  $Y$  beyond  $Y$ ’s own past.” Stance Granger-causes returns if the lagged stance coefficients are jointly nonzero in the return equation, i.e.  $H_0 : a_1^{(2,1)} = \dots = a_p^{(2,1)} = 0$  is rejected. We test this with a direct OLS regression of each variable on  $p$  lags of both variables, using a joint Wald ( $F$ ) test and Newey–West HAC-robust standard errors (`maxlags = p`) to neutralize the heteroscedasticity and autocorrelation found

TABLE IV  
FIRM-LEVEL GRANGER CAUSALITY, FULL SAMPLE (HAC  $p$ -VALUES)

| Firm                                                   | Lag | Bias→Ret | Ret→Bias |
|--------------------------------------------------------|-----|----------|----------|
| GS                                                     | 5   | 0.757    | 0.012    |
| MS                                                     | 3   | 0.058    | 0.193    |
| WFC                                                    | 5   | 0.023    | 0.307    |
| <i>Other 12 firms: no direction significant at 5%.</i> |     |          |          |

in Stage 7. Table IV reports the firm-level full-sample  $p$ -values. Causality is sparse in the full sample, as expected if the relationship is regime-dependent: stance Granger-causes returns for Wells Fargo ( $p = 0.023$ ) and weakly for Morgan Stanley ( $p = 0.058$ ), and the reverse direction (returns to bias) is significant for Goldman Sachs ( $p = 0.012$ ).

The more informative test splits each firm at its empirical bias break and re-estimates causality on each side. Several firms that are quiet in the full sample show post-break causality that is absent pre-break: Wells Fargo (bias→returns  $p = 0.013$  post versus 0.345 pre) and Morgan Stanley (bias→returns  $p = 0.052$  post) are the clearest cases, with feedback (returns→bias) emerging post-break for Boeing, Goldman Sachs, and Morgan Stanley. This pre/post asymmetry is the firm-level analogue of the structural shift the regressions probe: the media–market link is not constant but switches on after the break for a subset of firms.

#### G. Panel VAR

Firm-by-firm tests are underpowered because each uses only one short series. We therefore pool the cross-section into a panel VAR, which is the design the study most relies on for its dynamic conclusion. Unobserved firm heterogeneity is removed with the Arellano–Bover/Helmert transformation (forward mean-differencing), which subtracts from each observation the mean of all its *future* values within the firm,

$$\tilde{y}_{it} = c_{it} \left( y_{it} - \frac{1}{T_{it}} \sum_{s>t} y_{is} \right), \quad (8)$$

with a scaling factor  $c_{it}$  that equalizes the variance of the transformed errors. Unlike ordinary within-demeaning, this transformation preserves the orthogonality between the transformed variables and the lagged regressors, avoiding the look-ahead bias (Nickell bias) that contaminates dynamic fixed-effects panels. The endogenous variables are again  $\Delta \text{bias}$  and the weekly log return; we additionally partial out the exogenous controls  $\text{Post}_t$ , VIX, the S&P 500 weekly log return, and the article count  $N_{\text{articles}}$ , so that any detected causality cannot be attributed to market-wide volatility or coverage volume. With  $p = 3$  lags and HAC-robust standard errors, the pooled model is estimated by OLS on the transformed, constant-free data, and Granger directions are tested with joint Wald statistics over the full, pre-break, and post-break samples. Table V reports the results.

Pooling buys power but, even so, neither Granger direction is significant at conventional levels in any subsample once VIX, the index return, and article volume are controlled.

TABLE V  
PANEL VAR GRANGER CAUSALITY (15 FIRMS, HELMERT-TRANSFORMED, HAC)

| Sample     | $N$  | Bias→Ret | Ret→Bias |
|------------|------|----------|----------|
| Full       | 7493 | 0.976    | 0.200    |
| Pre-break  | 4168 | 0.245    | 0.801    |
| Post-break | 3265 | 0.617    | 0.187    |

The honest reading is that the firm-level post-break signals for Wells Fargo and Morgan Stanley do not aggregate into a uniform panel-wide media-to-returns channel: the dynamic link, where it exists, is firm-specific and heterogeneous rather than a market-wide regularity. This strengthens rather than weakens the contribution, because the panel VAR is exactly the test that would manufacture a spurious aggregate effect if the controls were omitted, and it declines to do so.

#### H. Limitations of the VAR Pipeline

Several caveats bound the dynamic findings. A weekly frequency, chosen for coverage, may be too coarse to capture the intraday or daily horizon over which news is plausibly priced, so a true fast media-to-price channel could be aliased away by aggregation. The single-break Bai–Perron design dates exactly one change point per series; firms with multiple regime shifts are only partially described, and the break date is estimated with sampling error that the subsample split treats as known. The bivariate VAR omits other drivers of both stance and returns (earnings surprises, analyst actions, sector shocks), so the controls in the panel model mitigate but do not eliminate omitted-variable concerns. Heavy-tailed, volatility-clustered residuals mean small-sample inference leans on the HAC correction and asymptotics rather than exact distributions. Finally, Granger causality is predictive, not structural: a significant test says past stance forecasts returns, not that stance mechanically moves prices, and the firm-specific nature of the surviving signals counsels caution against a universal causal claim.

#### VII. IDENTIFICATION, VALIDITY, AND ROBUSTNESS

We address three main threats. First, macro shocks may move both coverage and prices; we control with index returns and VIX. Second, article volume differs widely across firms; we include volume covariates and run trimmed samples. Third, parallel trends may not hold; we use pre-trend checks, placebo cutoffs, and SCDiD when needed.

Robustness checks include alternate stance thresholds (0.9, 0.6, 0.0), alternative aggregation (mean vs median), lagged stance controls, sector controls, and quantile regressions to test for tail effects in returns.

#### VIII. CONCLUSION AND NEXT STEPS

This draft provides a full narrative of the project to date: why the 2020 shock matters, how we build a target-dependent stance signal, and how we test for structural changes and market effects. Next steps include finalizing the firm panel,

completing the relevance classifier evaluation, computing the daily stance series, and running the full RQ1–RQ3 specifications with robustness tables and figures.

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